

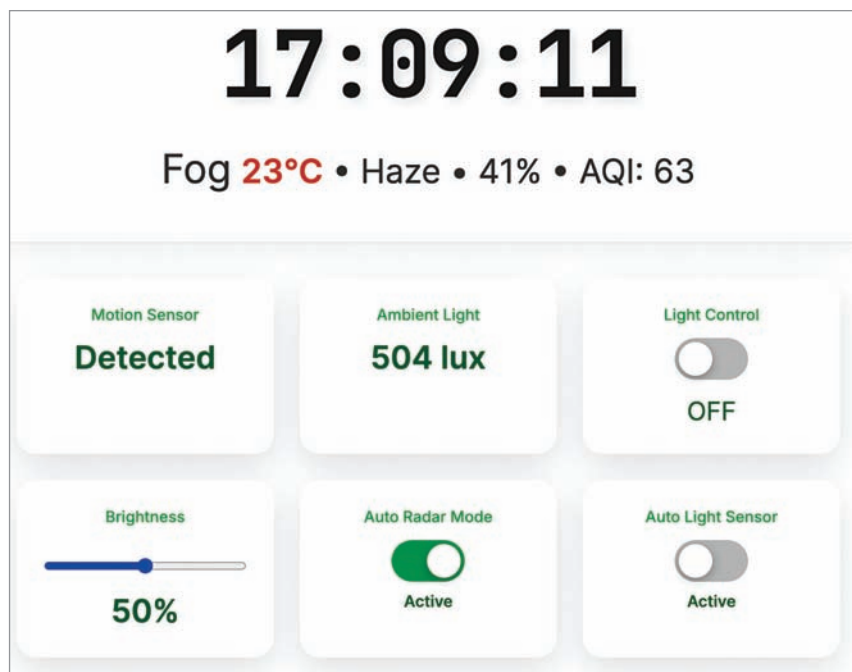


Fig. 3: UI of web dashboard

```

1  #include <WiFi.h>
2  #include <WebServer.h>
3  #include <ArduinoJson.h>
4  #include <WiFiUdp.h>
5  #include <NTPClient.h>
6  #include <HTTPClient.h>
7
8  // ===== PINS =====
9  #define RCWL_PIN      1
10 #define LIGHT_SENSOR_PIN 2
11 #define OPTO_PIN      3
12
13 // ===== WIFI =====
14 const char* ssid = "YourWiFiSSID"; // CHANGE THIS
15 const char* password = "YourWiFiPassword"; // CHANGE THIS
16 const char* ap_ssid = "IndusBoardSSR";
17 const char* ap_password = "12345678";
18
19 // ===== TIME & WEATHER (FREE) =====
20 WiFiUDP ntpUDP;
21 NTPClient timeClient(ntpUDP, "pool.ntp.org", 19800, 60000); // India Time
22
23 String current_time = "Connecting...";
24 String current_temp = "--°C";
25 String current_weather = "Loading...";
26 String current_aqi = "AQI: --";
27
28 unsigned long last_update = 0;
29 const unsigned long UPDATE_INTERVAL = 600000; // 10 min
30
31 // ===== GLOBALS =====
32 WebServer server(80);
33 bool wifi_connected = false;
34
35 bool auto_radar = true;
36 bool auto_light = false;
37 unsigned long motion_start = 0;
38 const unsigned long AUTO_OFF_TIME = 30000;
39

```

Fig. 4: Code snippet - Configure AP and STA, SSID and Password

The RCWL-0516 microwave radar module receives power from the 3.3V and GND pins on the board, while its OUT pin is routed via a solder jumper (JP1) for signal selection. This jumper routes the radar's detection signal directly to a selected I/O pin of the IndusBoard controller for processing. The unused pins VIN and CDS remain open in this configuration, as they are not required for the current design.

The radar detection signal can optionally drive the solid-state relay optocoupler (VO2223A) via a current-limiting resistor (R1). When activated, the optocoupler triggers its internal triac, enabling AC load switching. A snubber network consisting of resistor R3 and capacitor C3 is connected across the triac terminals to ensure reliable operation and protect the device from voltage spikes caused by inductive loads.

A 2-terminal connector is provided for attaching the AC bulb. The live wire (L) passes through the optocoupler output, while the neutral wire (N) connects directly to the bulb. Additional flexibility is provided through a second jumper (JP2), which enables the microcontroller to control the relay output or monitor the AC load state through another IO pin. The FPC connector accommodates an OV2640 camera module, providing scope for future feature expansion and enhanced system capability.

Software

The code is written using the Arduino IDE, and the ESP32 board package must be installed first. The program includes an additional configuration section that enables the device to operate in STA mode, AP mode, or both simultaneously. This behaviour is selected in the code by assigning 0 to STA mode, 1 to AP mode, and 2 to dual-mode operation, as shown in Fig. 3.

A user interface (UI) is created to control and interact with the device. The UI also includes a feature that